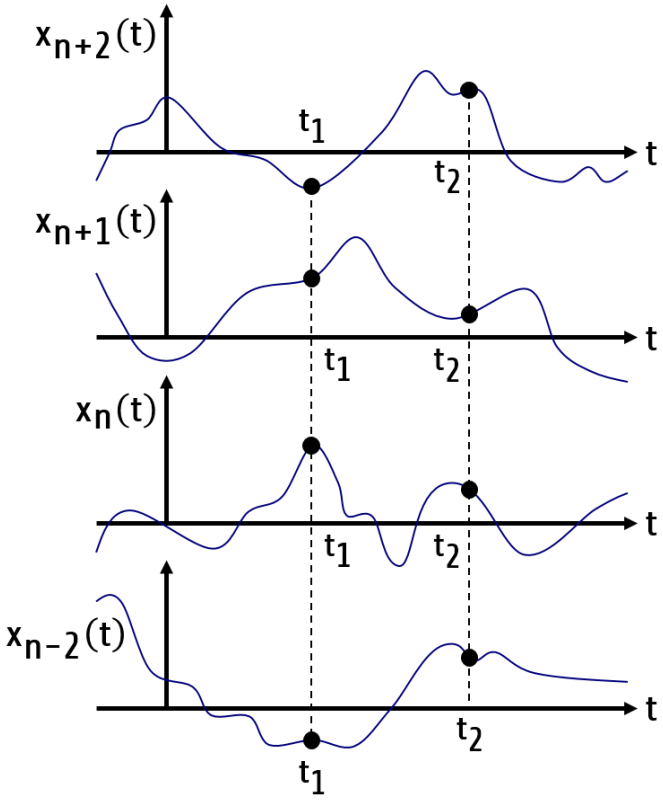
Random Process



Random Process Concept

- Random variable: assigns a function $x(s)$ to every outcome s.
- Random process: assigns a time function $x(t,s)$ to every outcome s
- Random process $X(t,s)$ = family of all such functions





Random Process Concept

- Each member time function = sample function, ensemble member, realization of the process.
- X(t,s) can denote the following quantities:
  1. $X(t,s) = \{X(t,s_i | s_i \in S\} = \{x_1(t), x_2(t), \dots\}$, a collection of functions of time
  2. $X(t,s_i) = x_i(t)$, a specific member function
  3. $X(t_0,s) = \{X(t_0,s_i) |\ s_i\ \in S\ \} = \{x_1(t_0\},\ x_2(t_0), \dots\}$, a collection of the numerical values of the member functions at t = t_0, i.e., a random variable
  4. $X(t_0,s_i) = x_i(t_0)$, numerical value of the ith member function at t = t_0.



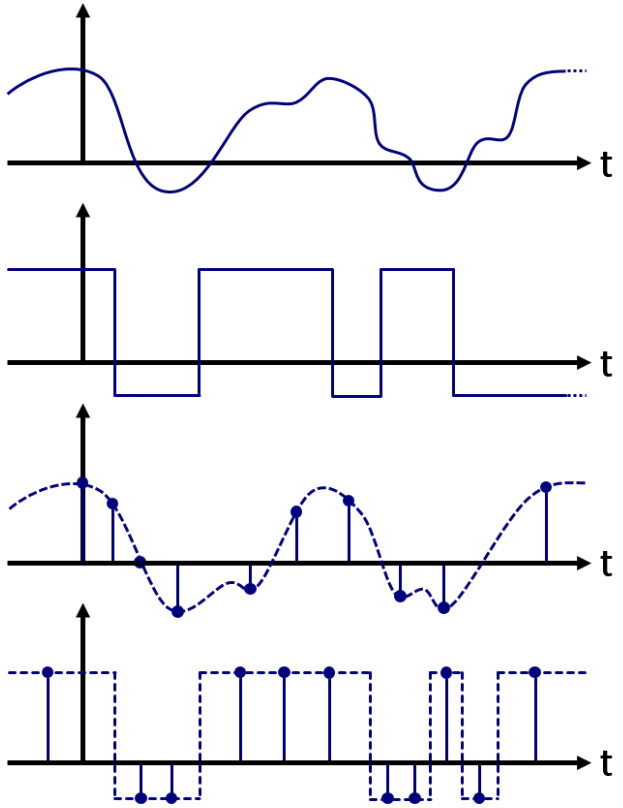
Classification of Random Processes

	t	
X(t)	**Continuous**	**Discrete**
Continuous	Continuous random process	Continuous random sequence
Discrete	Discrete random process	Discrete random sequence



Classification of Random Processes

	t	
X(t)	**Continuous**	**Discrete**
Continuous	Continuous random process	Continuous random sequence
Discrete	Discrete random process	Discrete random sequence





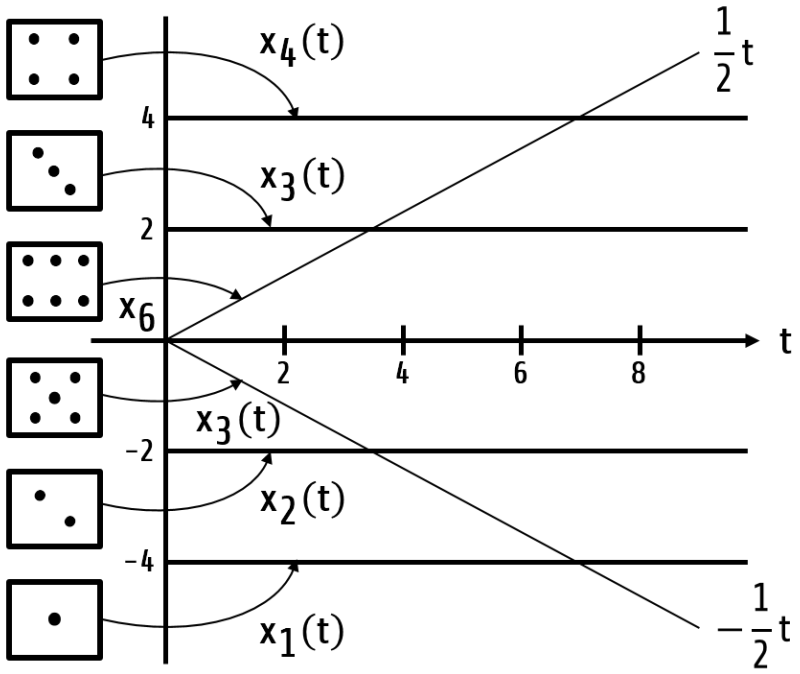
Example

- Consider a random experiment that consists of tossing a die at $t = 0$ and observing the number of dots showing on the top face. For each outcome, let us arbitrarily assign the following functions of time t, $0 \leq t \leq \infty$.



Example

Outcome	Waveform
1	$X_1(t) = -4$
2	$x_2(t) = -2$
3	$x_3(t) = +2$
4	$x_4(t) = +4$
5	$x_5(t) = -\frac{t}{2}$
6	$x_6(t) = +\frac{t}{2}$

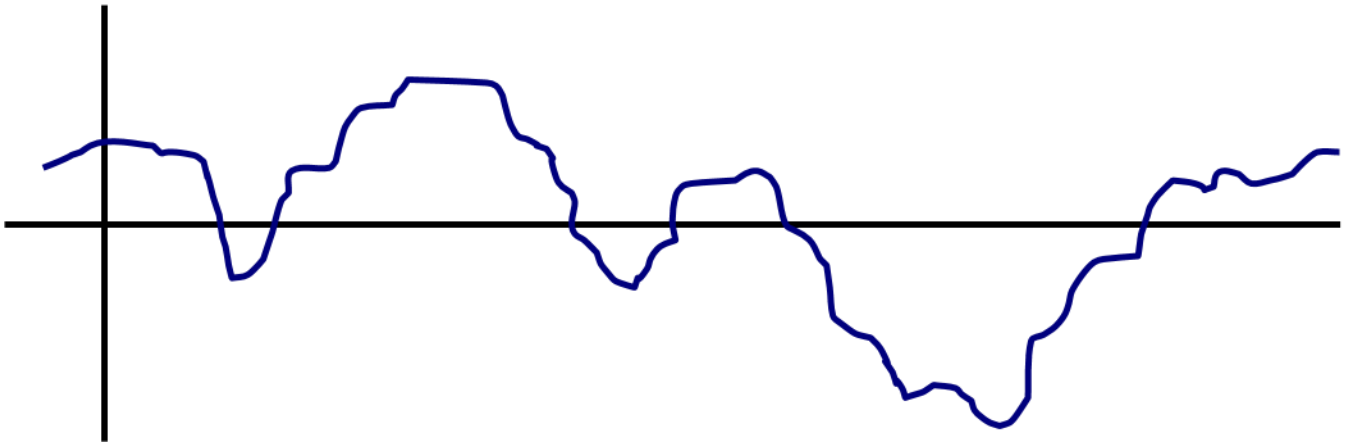


The set of waveforms $\{x_1(t), x_2(t), \ldots, x_6(t)\}$, represents this random process and are called the ensemble.



Classification of Random Processes

- Nondeterministic process – if future values of any sample function cannot be predicted exactly from observed past values
  - Example:





Classification of Random Processes

- Deterministic process – if future values of any sample function can be predicted exactly from observed past values
  - Example: $X(t) = A\cos(\omega_0 t + \theta)$, where A, θ, ω_0 may be random variables



Methods of Description

1. Joint distributions of Time Samples
2. Analytical description using random variables
3. Average values



Joint Distribution

- Let $x_1, x_2, \ldots, x_k$ be the k random variables obtained by sampling the random process $x(t, \xi)$ at the times $t_1, t_2, \ldots, t_k$:

$$x_1 = x(t_1, \xi),\ x_2 = x(t_2, \xi), \ldots, x_k = x(t_k, \xi)$$

- The joint behavior of the random process at these k time instants is specified by the joint cumulative distribution of the vector random variable $(x_1, x_2, \ldots, x_k)$.



Joint Distribution

- A stochastic/random process is specified by the collection of kth order joint cumulative distribution functions:

$$F_{x_1 x_2 \ldots x_k}(x_1, x_2, \ldots, x_k) =$$
$$P[X_1 \leq x_1, X_2 \leq x_2, \ldots, X_k \leq x_k]$$

for any k and any choice of sampling instants t_1,..., t_k.



Joint Distribution

- If the stochastic process is discrete-valued, then a collection of pmf's can be used to specify the stochastic process:

$$P_{x_1 x_2 \ldots x_k}(x_1, x_2, \ldots, x_k) =$$
$$P[X_1 = x_1, X_2 = x_2, \ldots, X_k = x_k]$$

- If the stochastic process is continuous-valued, then a collection of pdf's can be used instead:

$$f(x_1, \ldots, x_k)$$



Joint Distribution

- At least one k-variate distribution function is required for each value of k. However, the first-order distribution function(s) $P[x(t_1) \leq a_1]$ and the second-order distributed function(s) $P[x(t_1) \leq a_1, x(t_2) \leq a_2]$ are primarily used.
  - 1st order cdf: describes the instantaneous amplitude distribution of the process.
  - 2nd order cdf: tells something about the structure of the signal in the time-domain and thus the spectral content of the signal.



Joint Distribution

- While the joint distribution functions of a process can be derived from a description of the random experiment and the mapping, there is no technique for constructing member functions from joint distribution functions.



Analytical Description using Random Variables

- Example:
  - represent the ensemble of a random process Y(t) of the form $Y(t) = A\cos(10^3 t + \theta)$ where A, θ are random variables



Average Values

- The moments of time samples of a random process can be used to partially specify the random process because they summarize the information contained in the joint cdf's.
- The mean of a random process X(t) is defined by

$$\mu_x(t) \triangleq E[X(t)] = \int_{-\infty}^{\infty} x\, f_{x(t)}(x)\, dx$$

where $f_{x(t)}(x)$ is the pdf of X(t).



Average Values

- In general, $\mu_x(t)$, is a function of time. Trends in the behavior of x(t) are reflected in the variation of $\mu_x(t)$with time.



Average Values

- The autocorrelation of a random process $x(t)$ is defined as the joint moment of $X(t_1)$ and $X(t_2)$:

$$R_{XX}(t_1, t_2) \triangleq E[X(t_1)X(t_2)]$$

$$= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 \, f_{x(t_1)x(t_2)}(x_1, x_2) dx_1 dx_2$$

where $f_{x(t_1)x(t_2)}(x, y)$ is the second-order pdf of $X(t)$. In general, the autocorrelation is a function of t_1 and t_2.



Average Values

- The autocovariance of a random process x(t) is defined as the covariance of $X(t_1)$ and $X(t_2)$:

$$C_{XX}(t_1, t_2) \triangleq E\{[X(t_1) - \mu_x(t_1)][X(t_2) - \mu_x(t_2)]\}$$
$$= R_{XX}(t_1, t_2) - \mu_x(t_1)\mu_x(t_2)$$



Average Values

- The variance of x(t) can be obtained from $C_{XX}(t_1, t_2)$:

$$var[x(t)] = E\{[X(t) - \mu_x(t)]^2\} = C_{XX}(t, t)$$

- The correlation coefficient of x(t) is defined as the correlation coefficient of $x(t_1)$ and $x(t_2)$:

$$P_{xx}(t_1, t_2) = \frac{C_{xx}(t_1, t_2)}{\sqrt{C_{xx}(t_1, t_1)}\sqrt{C_{xx}(t_2, t_2)}}$$

(autocorrelation coefficient)



Average Values

- Example: Previous example.

$$\mu_x(t) = E[x(t)] = \frac{1}{6}\sum_{i=1}^{6} x_i(t) = 0$$

$$R_{xx}(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$= \frac{1}{6}\sum_{i=1}^{6} x_i(t_1)x_i(t_2)$$

$$= \frac{1}{6}\left\{16 + 4 + 4 + 16 + \frac{1}{4}t_1t_2 + \frac{1}{4}t_1t_2\right\}$$

$$= \frac{1}{6}\left\{40 + \frac{1}{2}t_1t_2\right\}$$



Average Values

- Example: Previous example.

$$C_{xx}(t_1, t_2) = R_{xx}(t_1, t_2)$$

$$P_{xx}(t_1, t_2) = \frac{40 + \frac{1}{2} t_1 t_2}{\sqrt{(40 + \frac{1}{2} t_1^2)(40 + \frac{1}{2} t_2^2)}}$$



Average Values

- Example:

A random process has the functional form:

$$X(t) = A\cos(100t + \theta)$$

where A is a normal random variable with a mean of 0 and variance of 1, and θ is uniformly distributed in the interval $[-\pi, \pi]$. Assuming A and θ are independent random variables, find $\mu_x(t)$ and $R_{xx}(t, t+\tau)$



Average Values

- Example:

$$\mu_{x(t)} = E[A]E[\cos(100t + \theta)] = 0$$

$$\begin{aligned} R_{xx(t,t+\tau)} &= E[x(t_1)x(t_2), t_1 = t, t_2 = t + \tau \\ &= E[A\cos(100t + \theta)\, A\cos(100t + 100\tau + \theta)] \\ &= E[\frac{A^2}{2}[\cos(100t) + \cos(200t + 100\tau + 2\theta)]] \\ &= \frac{A^2}{2}\cos(100t) + \frac{A^2}{2}E[\cos(200t + 100\tau + 2\theta)] \\ &= \frac{A^2}{2}\cos(100\tau) \end{aligned}$$

Since $E\{\cos(200t + 100t + 2\theta)\} = 0$



Average Values

- Example: Sinusoid with Random Amplitude.

Let $x(t) = A\cos 2\pi t$ where A is a random variable

$$\mu_x(t) = E[A\cos 2\pi t] = E[A]\cos 2\pi t$$

$$\begin{aligned} R_{xx}(t_1, t_2) &= E[A\cos 2\pi t_1 \;\; A\cos 2\pi t_2] \\ &= E[A^2]\cos 2\pi t_2 \cos 2\pi t_1 \end{aligned}$$

$$\begin{aligned} C_{xx}(t_1, t_2) &= R_{xx}(t_1, t_2) - \mu_x(t_1)\mu_x(t_2) \\ &= \{E[A^2] - E[A]^2\}\cos 2\pi t_1 \cos 2\pi t_2 \\ &= var[A]\cos 2\pi t_1 \cos 2\pi t_2 \end{aligned}$$



Average Values

- Example: Sinusoid with Random Phase.

Let $x(t) = \cos(\omega t + \theta)$ where θ is a random variable

$$\mu_x(t) = E[\cos(\omega t + \theta)] = \frac{1}{2\pi}\int_{-\pi}^{\pi} \cos(\omega t + x)\, dx = 0$$

$$\begin{aligned} C_{xx}(t_1, t_2) &= R_{xx}(t_1, t_2) \\ &= E[\cos(\omega t_1 + \theta)\cos(\omega t_2 + \theta)] \\ &= \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{1}{2}\{\cos\big(\omega(t_1 - t_2)\big) + \cos(\omega(t_1 + t_2) + 2x)\}dx \\ &= \frac{1}{2}\cos(\omega(t_1 - t_2)) \end{aligned}$$

where $\cos(a)\cos(b) = \frac{1}{2}\cos(a + b) + \frac{1}{2}\cos(a - b)$



Average Values

- Example: Sinusoid with Random Phase.

Let $x(t) = \cos(\omega t + \theta)$ where θ is a random variable

Note: $\mu_x(t)$is a constant. $C_{xx}(t_1, t_2)$ depends only on $|t_1 - t_2|$



Two or More Random Processes

- Consider two random processes X(t) and Y(t) whose joint distribution function is denoted by

$$P[x(t_1) \leq x_1, \ldots, x(t_n) \leq x_n, Y(t_1') \leq y_1, .., Y(t_m') \leq y_m]$$

- The cross-correlation of X(t) and Y(t) is defined by

$$R_{XY}(t_1, t_2) = E[X(t_1)Y(t_2)]$$

- The processes X(t) and Y(t) are said to be orthogonal if

$$R_{XY}(t_1, t_2) = 0, \qquad \forall\, t_1, t_2$$



Two or More Random Processes

- The cross-covariance of X(t) and Y(t) is defined by

$$C_{XY}(t_1, t_2) = E\{[X(t_1) - \mu_X(t_1)][Y(t_2) - \mu_Y(t_2)]\}$$
$$= R_{XY}(t_1, t_2) - \mu_x(t_1)\mu_Y(t_2)$$

- The processes X(t) and Y(t) are said to be uncorrelated if

$$C_{XY}(t_1, t_2) = 0, \qquad \forall\, t_1, t_2$$



Two or More Random Processes

- Correlation Coefficient (cross-correlation coefficient):

$$P_{XY}(t_1, t_2) \triangleq \frac{C_{XY}(t_1, t_2)}{\sqrt{C_{XX}(t_1, t_1) C_{YY}(t_2, t_2)}}$$

*For random sequences, replace t_1 and t_2 with n_1 and n_2



Two or More Random Processes

- Equality of two random processes will mean that their respective member functions are identical for each outcome $\lambda \in S$. Note that equality also implies that the processes are defined on the same random experiment.



Two or More Random Processes

- Random processes $X(t)$ and $Y(t)$ are independent if

$$P\begin{bmatrix} X(t_1) \le x_1, \ldots, X(t_n) \le x_n, \\ Y(t_1') \le y_1, .., Y(t_m') \le y_m \end{bmatrix}$$
$$= P[X(t_1) \le x_1, \ldots, X(t_n) \le x_n,] \times$$
$$P[Y(t_1') \le y_1, .., Y(t_m') \le y_m]$$

$$\forall\ n, m\ and\ t_1, \ldots, t_n, t_1', \ldots, t_m'$$



Two or More Random Processes

- Example:

Let $X(t) = \cos(\omega t + \theta)$

$Y(t) = \sin(\omega t + \theta)$

θ = random variable uniformly distributed in $[-\pi, \pi]$

Find the cross-covariance of X(t) and Y(t).



Two or More Random Processes

- Example:

From previous examples we know that X(t) and Y(t) are zero-mean. Cross-covariance = cross-correlation.

$$R_{XY}(t_1, t_2) = E[\cos(\omega t_1 + \theta)\sin(\omega t_2 + \theta)]$$
$$= E[-\frac{1}{2}\sin(\omega(t_1 - t_2)) + \frac{1}{2}\sin(\omega(t_1 + t_2) + 2\theta)]$$
$$= -\frac{1}{2}\sin(\omega(t_1 - t_2))$$

since $E[\sin(\omega(t_1 + t_2) + 2\theta)] = 0$



Two or More Random Processes

- Example:

Suppose we observe a process $\mathrm{Y(t)}$, which consists of a desired signal $\mathrm{X(t)}$ plus noise $\mathrm{N(t)}$:

$$Y(t) = X(t) + N(t)$$

Find the cross-correlation between the observed signal and the desired signal assuming that $X(t)$ and $N(t)$ are independent random processes.



Two or More Random Processes

- Example:

Find the cross-correlation between the observed signal and the desired signal assuming that $X(t)$ and $N(t)$ are independent random processes.

$$\begin{aligned} R_{XY}(t_1, t_2) &= E[X(t_1)Y(t_2)] \\ &= E[X(t_1)\{X(t_2) + N(t_2)\}] \\ &= E[X(t_1)X(t_2)] + E[X(t_1)N(t_2)] \\ &= R_{XX}(t_1, t_2) + E[X(t_1)]E[N(t_2)] \\ &= R_{XX}(t_1, t_2) + \mu_x(t_1)\mu_N(t_2) \end{aligned}$$



Examples of Discrete-Time Random Processes

- iid Random Processes

Let X_n = a discrete-time random process consisting of a sequence of iid random variables with common cdf $F_X(x)$, mean μ, and variance σ^2

X_n = the iid random process.



Examples of Discrete-Time Random Processes

- iid Random Processes

The joint cdf for any time instants $\mathrm{n}_1, \mathrm{n}_2, \ldots, \mathrm{n}_\mathrm{k}$ is given by

$$\begin{aligned} F_{X_1 \ldots X_k}(x_1, x_2, \ldots, x_k) \\ &= P[X_1 \le x_1, X_2 \le x_2, \ldots, X_k \le x_k] \\ &= F_X(x_1) F_X(x_2) \ldots F_X(x_k) \end{aligned}$$

where $\mathrm{X_k} = \mathrm{X_{nk}}$



Examples of Discrete-Time Random Processes

- iid Random Processes

→ If X_n is discrete-valued, the joint pmf factors into the product of individual pmf's

→ If X_n is continuous-valued, the joint pdf factors into the product of individual pdf's



Examples of Discrete-Time Random Processes

- iid Random Processes

Mean of an iid process:

$$\mu_x(n) = E[X_n] = \mu, \text{ for all n}$$

→ The mean is constant



Examples of Discrete-Time Random Processes

- iid Random Processes

Autocovariance Function:

if $n_1 \neq n_2$,

$$C_{xx}(n_1, n_2) = E\left[\left(X_{n_1} - \mu\right)\left(X_{n_2} - \mu\right)\right]$$
$$= E\left[\left(X_{n_1} - \mu\right)\right]E\left[\left(X_{n_2} - \mu\right)\right]$$
$$= 0$$

since X_{n_1} and X_{n_2} are independent random variables



Examples of Discrete-Time Random Processes

- iid Random Processes

Autocovariance Function:

if $\mathrm{n}_1 = \mathrm{n}_2 = \mathrm{n}$,

$$C_{xx}(n_1, n_2) = E[(X_n - \mu)^2] = \sigma^2$$

in general

$$C_{xx}(n_1, n_2) = \sigma^2 \, \delta_{n_1 n_2}$$

where $\delta_{\mathrm{n}_1\mathrm{n}_2} = 1$ if $\mathrm{n}_1 = \mathrm{n}_2$ and

$\delta_{\mathrm{n}_1\mathrm{n}_2} = 0$ if otherwise



Examples of Discrete-Time Random Processes

- iid Random Processes

Autocorrelation Function:

$$R_{XX}(n_1, n_2) = C_{XX(n_1,n_2)} + \mu^2$$



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

Let I_n = a sequence of independent Bernoulli random variables

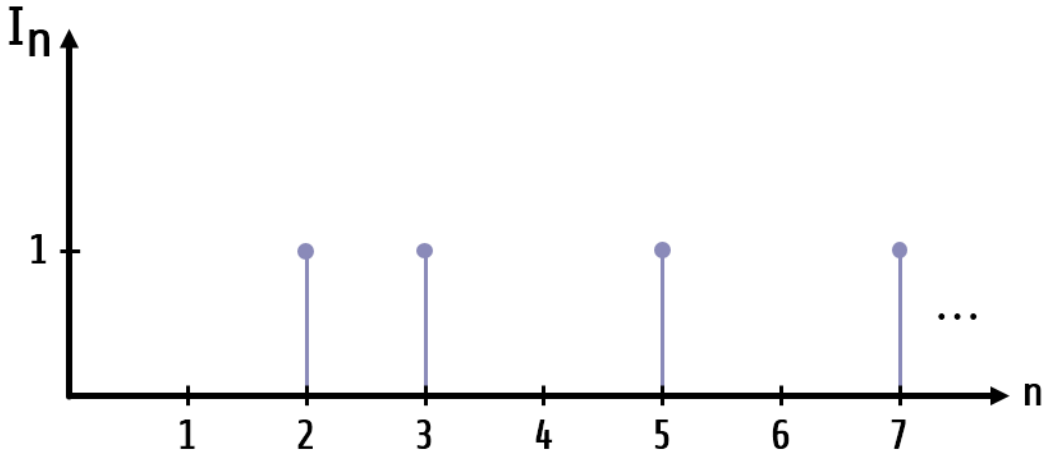
= an iid random process taking on values from the set $\{0,1\}$.



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

A realization of the process:



e.g. I_n = an indicator function for the event "a light bulb fails and is replaced on day n"



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

Since $\mathrm{I_n}$ is a Bernoulli random variable,

$$\mu_I(n) = p \quad var[I_n] = p(1-p)$$

The probability that the 1st 4 bits in the sequence are 1001 is

$$\begin{aligned} & P[I_1 = 1, I_2 = 0, I_3 = 0, I_4 = 1] \\ &= P[I_1 = 1]P[I_2 = 0]P[I_3 = 0]P[I_4 = 1] \\ &= p^2(1-p)^2 \end{aligned}$$



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

The probability that the 2nd bit is 0 and the 7th is 1 is

$$P[I_2 = 0, I_7 = 1] = P[I_2 = 0]P[I_7 = 1] = p(1-p)$$



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

Let $\mathrm{D_n} = 2\mathrm{I_n} - 1$, where $\mathrm{I_n}$ is the Bernoulli random process. Then

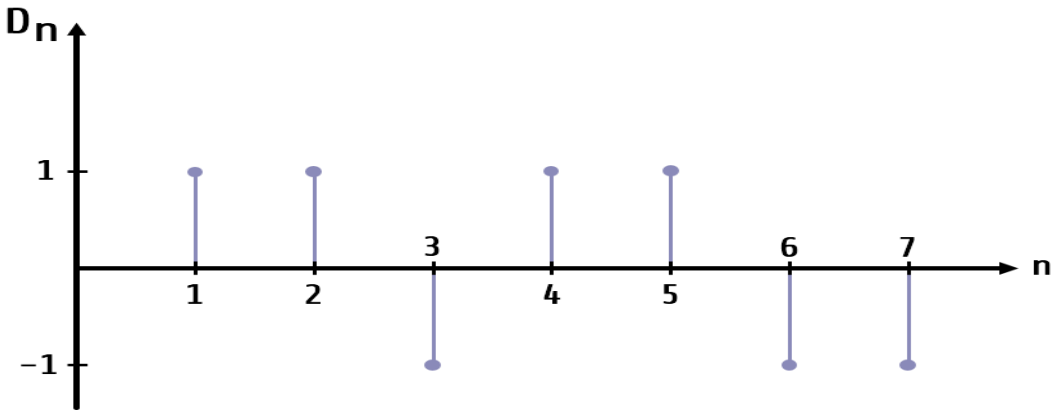
$$D_n = \begin{cases} 1, & if\ I_n = 1 \\ -1, & if\ I_n = 0 \end{cases}$$



Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

Let D_n represent the change in position of a particle that moves along a straight line in jumps of +/- 1 every time unit.





Examples of Discrete-Time Random Processes

- Example: Bernoulli Random Process

Let $\mathrm{D_n}$ represent the change in position of a particle that moves along a straight line in jumps of +/- 1 every time unit.

$$\text{mean: } \mu_D(n) = E[D_n] = E[2I_n - 1]$$
$$= 2E[I_n] - 1 = 2p - 1$$
$$\text{variance: } var(D_n) = var[2I_n - 1]$$
$$= 2^2 var[I_n] = 4p(1-p)$$



Examples of Discrete-Time Random Processes

- Sum Processes: The Binomial Counting and Random Walk Processes

Let S_n = sum of a sequence of iid random variables

$= X_1 + X_2 + \cdots + X_n, n = 1, 2, \ldots$

$= S_{n-1} + X_n$, where $S_0 = 0$

S_n = the sum process



Examples of Discrete-Time Random Processes

- Example: Binomial Counting Process

Let I_i = sequence of independent Bernoulli random variables

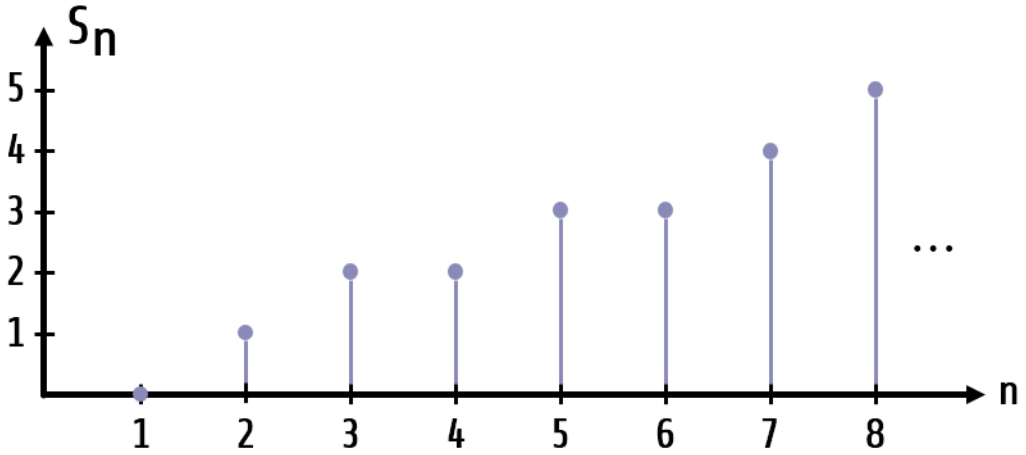
S_n = counting process that gives the number of successes in the 1st n Bernoulli trials



Examples of Discrete-Time Random Processes

- Example: Binomial Counting Process

Sample function for S_n





Wiener Process or Brownian Motion

Suppose that the symmetric random walk process (i.e., $p = {}^{1}/_{2}$) takes steps of magnitude h every δ seconds.

We obtain a continuous-time process by letting $X_\delta(t)$ be the accumulated sum up to time t. At time t, the process will have taken $n = {}^{t}/_{\delta}$ jumps, so it is equal to

$$X_\delta(t) = h(D_1 + D_2 + \cdots + D_{t/\delta} = hS_n$$



Wiener Process or Brownian Motion

Mean and variance of $X_\delta(t)$:

$$E[X_\delta(t)] = hE[S_n] = 0$$
$$Var[X_S(t)] = h^2 n\, Var[D_n] = h^2 n$$

Where we used the fact that $Var[D_n] = 4p(1-p) = 1$ since $p = {}^1/_2$

Suppose we let $\delta \to 0$ and $h \to 0$ with $h = \sqrt{\alpha\delta}$ and let $X(t)$ be the resulting process. $X(t)$ has

$$E[X(t)] = 0$$
$$Var[X(t)] = \left(\sqrt{\alpha\delta}\right)^2 \left({}^t/_\delta\right) = \alpha t$$



Wiener Process or Brownian Motion

$X(t) =$ continuous-time process that begins at the origin, has zero mean for all time, but has a variance that increases linearly with time.

$=$ Brownian motion r.p. (used to model the motion of particles suspended in a fluid that move under the rapid and random impact of neighboring particles)





Wiener Process or Brownian Motion



A $\delta \to 0$, $X(t)$ approaches the sum of an infinite no. of r.v.'s. The pdf of $X(t)$ approaches that of a Gaussian r.v. with zero mean and variance αt:

$$f_{X(t)}(x) = \frac{1}{\sqrt{2\pi\alpha t}} E^{-x^2/2\alpha t}$$

Autocovariance

$$C_{xx}(t_1, t_2) = \alpha \min(t_1, t_2)$$



Stationarity

- **Stationary process** – if its distribution functions or certain expected values are invariant with respect to a translation of the time axis.



First-Order Stationary Process

An r.p. is called stationary to order one if its first-order density function does not change with a shift in time origin:

$$f_X(X;t_1) = f_X(X_1;t_1+\Delta)$$
$$\forall\ t_1, real\ \Delta$$

Process mean value:

$$E[X(t)] = \mu_X = \text{constant}$$



Second-Order Stationary Process

A process is called stationary to order two if its second-order density function satisfies

$$f_X(X_1, X_2; t_1, t_2) = f_X(X_1, X_2; t_1 + \Delta, t_2 + \Delta)$$
$$\forall\ t_1, t_2, and\ \Delta$$



Second-Order Stationary Process

Notes: For a 2nd-order stationary process,

1. The 2nd order density function is a function of time differences $t_2 - t_1$ and not absolute time.
2. Also 1st-order stationary.
3. Auto correlation function is a function only of time differences and not absolute time.

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

let $\tau = t_2 - t_1$

$$R_{XX}(t_1, t_1 + \tau) = E[X(t_1)X(t_1 + \tau)]$$
$$= R_{XX}(\tau)$$



Wide-Sense Stationarity

A process $X(t)$ is said to be stationary in the wide sense (WSS or weakly stationary) if

$$E[X(t)] = \mu_X = \text{constant}$$
$$E[X(t)X(t+\tau)] = R_{XX}(\tau)$$

Two processes $X(t)$ and $Y(t)$ are jointly WSS if each process is WSS for all t, and

$$R_{XY}(t, t+\tau) = E[X(t)Y(t+\tau)] = R_{XY}(\tau)$$



Wide-Sense Stationarity

For random sequences, the conditions for WSS are

$$E[X(k)] = \mu_X$$
$$E[X(n)X(n+k)] = R_{XX}(k)$$

NOTE: A process stationary to order 2 is WSS. However, the converse is not necessarily true.



N-Order Stationarity

An r.p. is stationary to order N if its Nth-order density function is invariant to a time-origin shift:

$$f_X(X_1, \dots, X_N; t_1, \dots, t_N) = f_X(X_1, \dots, X_N; t_1 + \Delta, \dots, t_N + \Delta)$$
$$\forall t_1, \dots, t_N \; and \; \Delta$$

NOTE: Stationarity of order N implies stationarity to all orders $k \leq N$.



Strict-Sense Stationarity

A r.p. $X(t)$ is called time stationary or stationary in the strict sense (SSS) if all of the distribution functions describing the process are invariant under a translation of time:

$$P[X(t_1) \leq X_1, \dots, X(t_k) \leq X_k] = P[X(t_1 + \tau) \leq X_1, \dots, X(t_k + \tau) \leq X_k]$$

$$\forall\ t_1, t_2, \dots, t_k, t_1 + \tau, t_2 + \tau, \dots, t_k + \tau$$
$$\forall\ k = 1,2, \dots$$



Strict-Sense Stationarity

For a SSS process,

$$P[X(t) \leq X] = P[X(t+\tau) \leq X]$$

for any τ

$\rightarrow$ 1st-order distribution is independent of t

Similarly,

$$P[X(t_1) \leq X_1, X(t_2) \leq X_2] = P[X(t_1+\tau) \leq X_1, X(t_2+\tau) \leq X_2]$$

for any τ

$\rightarrow$ 2nd-order distribution is strictly a function of time diff. $t_2 = t_1$



Strict-Sense Stationarity

As a consequence,

$$E[X(t)] = \mu_X = \text{constant}$$
$$E[X(t_1)X(t_2)] = R_{XX}(t_2 - t_1)$$

Note: a r.p. with constant mean and autocorrelation that depends only on the time difference $t_2 - t_1$ need NOT even be 1st-order stationary.

- $X(t)$ and $Y(t)$ are jointly SSS if the jcdf are invariant under a translation of time.
- $Z(t) = X(t) + jY(t)$ is SSS if $X(t)$ and $Y(t)$ are jointly SSS.



Example

The r.p. $X(t)$ has the following sample functions:

$$X_1(t) = 5 \quad P_1 = {}^1/_6$$
$$X_2(t) = 3 \quad P_2 = {}^1/_6$$
$$X_3(t) = 1 \quad P_3 = {}^1/_6$$
$$X_4(t) = -1 \quad P_4 = {}^1/_6$$
$$X_5(t) = -3 \quad P_5 = {}^1/_6$$
$$X_6(t) = -5 \quad P_6 = {}^1/_6$$



Example

$X_1(t) = 5 \quad P_1 = 1/6$
$X_2(t) = 3 \quad P_2 = 1/6$
$X_3(t) = 1 \quad P_3 = 1/6$
$X_4(t) = -1 \quad P_4 = 1/6$
$X_5(t) = -3 \quad P_5 = 1/6$
$X_6(t) = -5 \quad P_6 = 1/6$

$$E[X(t)] = 0 \quad \text{for all } t$$

$$R_{XX}(t_1, t_2) = \frac{1}{6}(25 + 9 + 1 + 1 + 9 + 25) = \frac{70}{6}$$

Translation in time axis does not result in any change in any member function.

$$\therefore \boldsymbol{X(t)} \textbf{ is SSS}$$



Example

The r.p. $Y(t)$ has the following member functions:

$$Y_1(t) = 6 \qquad P_1 = 1/6$$
$$Y_2(t) = 3\sin t \qquad P_2 = 1/6$$
$$Y_3(t) = -3\sin t \qquad P_3 = 1/6$$
$$Y_4(t) = 3\cos t \qquad P_4 = 1/6$$
$$Y_5(t) = -3\cos t \qquad P_5 = 1/6$$
$$Y_6(t) = -6 \qquad P_6 = 1/6$$



Example

$$
\begin{aligned}
Y_1(t) &= 6 & P_1 &= 1/6 \\
Y_2(t) &= 3\sin t & P_2 &= 1/6 \\
Y_3(t) &= -3\sin t & P_3 &= 1/6 \\
Y_4(t) &= 3\cos t & P_4 &= 1/6 \\
Y_5(t) &= -3\cos t & P_5 &= 1/6 \\
Y_6(t) &= -6 & P_6 &= 1/6
\end{aligned}
$$

$$E[Y(t)] = 0$$

$$
\begin{aligned}
R_{YY}(t_1, t_2) &= \frac{1}{6}\begin{pmatrix} 25 + 9\sin t_1 \sin t_2 + 9\sin t_1 \sin t_2 \\ +9\cos t_1 \cos t_2 + 9\cos t_1 \cos t_2 + 36 \end{pmatrix} \\
&= \frac{1}{6}[72 + 18\ \cos(t_2 - t_1)] \\
&= R_{YY}(t_2 - t_1)
\end{aligned}
$$

$$\therefore \boldsymbol{Y(t)} \textbf{ is WSS}$$

$Y(t)$ is not SSS since the values that $Y(t)$ can have at $t = 0$ and $t = \pi/4$ are different and hence even the 1st-order distribution is not time invariant.



First-Order Markov Process

Definition: A r.p. $X(t)$ is called a first-order Markov (or Markoff) process if for all sequences of times $t_1 < t_2 < \cdots < t_k$ and $k = 1,2, \ldots$ we have

$$P[X(t_k) \leq X_k | X(t_{k-1}), \ldots, X(t_1)] = P[X(t_k) \leq X_k | X(t_{k-1})]$$

- Conditional probability distribution of $X(t_k)$ given all past values $X(t_1) = X_1, \ldots, X(t_{k-1}) = X_{k-1}$ depends only upon the most recent value $X(t_{k-1}) = X_{k-1}$.



Example 1

A binary-valued Markov sequence $X(n)$,

$$\mathrm{n} \in I = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$$

has the following joint probabilities:

$$P[X(n) = 0, X(n+1) = 0] = 0.2$$

$$P[X(n) = 0, X(n+1) = 1] = 0.2$$

$$P[X(n) = 1, X(n+1) = 0] = 0.2$$

$$P[X(n) = 1, X(n+1) = 1] = 0.4$$

Find $\mu_X(n), R_{XX}(n, n+1), R_{XX}(n, n+2), \ldots$, and show that the sequence is WSS.



Example 1

$$P[X(n)=0] = P[X(n)=0, X(n+1)=0] + P[X(n)=0, X(n+1)=1] = 0.4$$

$$P[X(n)=1] = 0.6$$

Hence $E[X(n)] = 0.6$ and $E[X(n)^2] = 0.6$

$$E[X(n)X(n+1)] = 1 \times P[X(n)=1, X(n+1)=1] = 0.4$$

$$E[X(n)X(n+1)] = 1 \times P[X(n)=1, X(n+2)=1]$$

$$= 1 \times P[X(n)=1, X(n+1)=1, X(n+2)=1] + 1 \times P[X(n)=1, X(n+1)=0, X(n+2)=1]$$

$$= 1 \times P[X(n)=1] P[, X(n+1)=1 | X(n)=1] \times P[X(n+2)=1 | X(n)=0, X(n+1)=0]$$



Example 1

Markov property implies that

$$P[X(n+2)=1|X(n)=1,X(n+1)=1]=P[X(n+2)=1|X(n+1)=1]$$
$$P[X(n+2)=1|X(n)=1,X(n+1)=0]=P[X(n+2)=1|X(n+1)=0]$$

Hence

$$E[X(n)X(n+2)]=0.6\left(\frac{0.4}{0.6}\right)\left(\frac{0.4}{0.6}\right)+0.6\left(\frac{0.2}{0.6}\right)\left(\frac{0.2}{0.6}\right)\approx 0.367$$

Thus,

$$\left.\begin{aligned} &E[X(n)]=0.6\\ &R_{XX}(n,n)=0.6\\ &R_{XX}(n,n+1)=0.4\\ &R_{XX}(n,n+2)=0.367 \end{aligned}\right\}\ \text{all independent of } n$$

We can show that $R_{XX}(n,n+k)$ will be independent of n.

$\therefore$ **WSS**



Example 2

A_i and B_i, $i = 1,2, \ldots, n$ is a set of $2n$ r.v.'s that are uncorrelated and have a joint Gaussian distribution with

$$E[A_i] = E[B_i] = 0 \text{ and } E\left[A_i^2\right] = E\left[B_i^2\right] = \sigma^2$$

Let

$$X(t) = \sum_{i=1}^{n} (A_i \cos \omega_i t + B_i \sin \omega_i t)$$

Show that $X(t)$ is a WSS Gaussian r.p.



Example 2

$$E[X(t)] = \sum_{i=1}^{n} [E(A_i)\cos\omega_i t + E(B_i)\sin\omega_i t] = 0$$

$$E[X(t)X(t+\tau)] = E\left\{ \begin{array}{c} \sum_{i=1}^{n}\sum_{j=1}^{n} (A_i \cos\omega_i t + B_i \sin\omega_i t) \\ \times \left[A_j \cos\omega_j (t+\tau) + B_j \sin\omega_j (t+\tau)\right] \end{array} \right\}$$

Since $E[A_i A_j], E[A_i B_i], E[A_i B_j], E[B_i B_j], i \neq j$ are all zero, we have

$$E[X(t)X(t+\tau)] = \sum_{i=1}^{n} \left[\begin{array}{l} E(A_i^2)\cos\omega_i t \cos\omega_i (t+\tau) \\ +E(B_i^2)\sin\omega_i t \sin\omega_i (t+\tau) \end{array} \right]$$

$$= \sigma^2 \sum_{i=1}^{n} \cos\omega_i \tau = R_{XX}(\tau)$$

$\therefore$ **$X(t)$ is WSS.**

